

Homework Data Visualization: Kittiong Wijaksanaboon

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Load library: tidyverse

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2     3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr       1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Data: mtcars, diamond

mtcars: The data was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

diamonds: A dataset containing the prices and other attributes of almost 54,000 diamonds. The variables are as follows:

```
mtcars %>% head(5)
```

```
##           mpg  cyl  disp  hp  drat    wt  qsec vs  am  gear  carb
## Mazda RX4      21.0    6  160  110  3.90  2.620  16.46  0   1    4    4
## Mazda RX4 Wag  21.0    6  160  110  3.90  2.875  17.02  0   1    4    4
## Datsun 710      22.8    4  108   93  3.85  2.320  18.61  1   1    4    1
## Hornet 4 Drive  21.4    6  258  110  3.08  3.215  19.44  1   0    3    1
## Hornet Sportabout 18.7    8  360  175  3.15  3.440  17.02  0   0    3    2
```

```
diamonds %>% head(5)
```

```
## # A tibble: 5 x 10
##   carat cut      color clarity depth table price      x      y      z
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal    E     SI2     61.5    55   326   3.95   3.98   2.43
## 2  0.21 Premium E     SI1     59.8    61   326   3.89   3.84   2.31
## 3  0.23 Good    E     VS1     56.9    65   327   4.05   4.07   2.31
## 4  0.29 Premium I     VS2     62.4    58   334   4.2    4.23   2.63
## 5  0.31 Good    J     SI2     63.3    58   335   4.34   4.35   2.75
```

Summary statistics data

```
summary(mtcars)
```

```
##      mpg      cyl      disp      hp
## Min.   :10.40  Min.   :4.000  Min.   : 71.1  Min.   : 52.0
## 1st Qu.:15.43  1st Qu.:4.000  1st Qu.:120.8  1st Qu.: 96.5
## Median :19.20  Median :6.000  Median :196.3  Median :123.0
## Mean   :20.09  Mean   :6.188  Mean   :230.7  Mean   :146.7
## 3rd Qu.:22.80  3rd Qu.:8.000  3rd Qu.:326.0  3rd Qu.:180.0
## Max.   :33.90  Max.   :8.000  Max.   :472.0  Max.   :335.0
##      drat      wt      qsec      vs
## Min.   :2.760  Min.   :1.513  Min.   :14.50  Min.   :0.0000
## 1st Qu.:3.080  1st Qu.:2.581  1st Qu.:16.89  1st Qu.:0.0000
## Median :3.695  Median :3.325  Median :17.71  Median :0.0000
## Mean   :3.597  Mean   :3.217  Mean   :17.85  Mean   :0.4375
## 3rd Qu.:3.920  3rd Qu.:3.610  3rd Qu.:18.90  3rd Qu.:1.0000
## Max.   :4.930  Max.   :5.424  Max.   :22.90  Max.   :1.0000
##      am      gear      carb
## Min.   :0.0000  Min.   :3.000  Min.   :1.000
## 1st Qu.:0.0000  1st Qu.:3.000  1st Qu.:2.000
## Median :0.0000  Median :4.000  Median :2.000
## Mean   :0.4062  Mean   :3.688  Mean   :2.812
## 3rd Qu.:1.0000  3rd Qu.:4.000  3rd Qu.:4.000
## Max.   :1.0000  Max.   :5.000  Max.   :8.000
```

```
summary(diamonds)
```

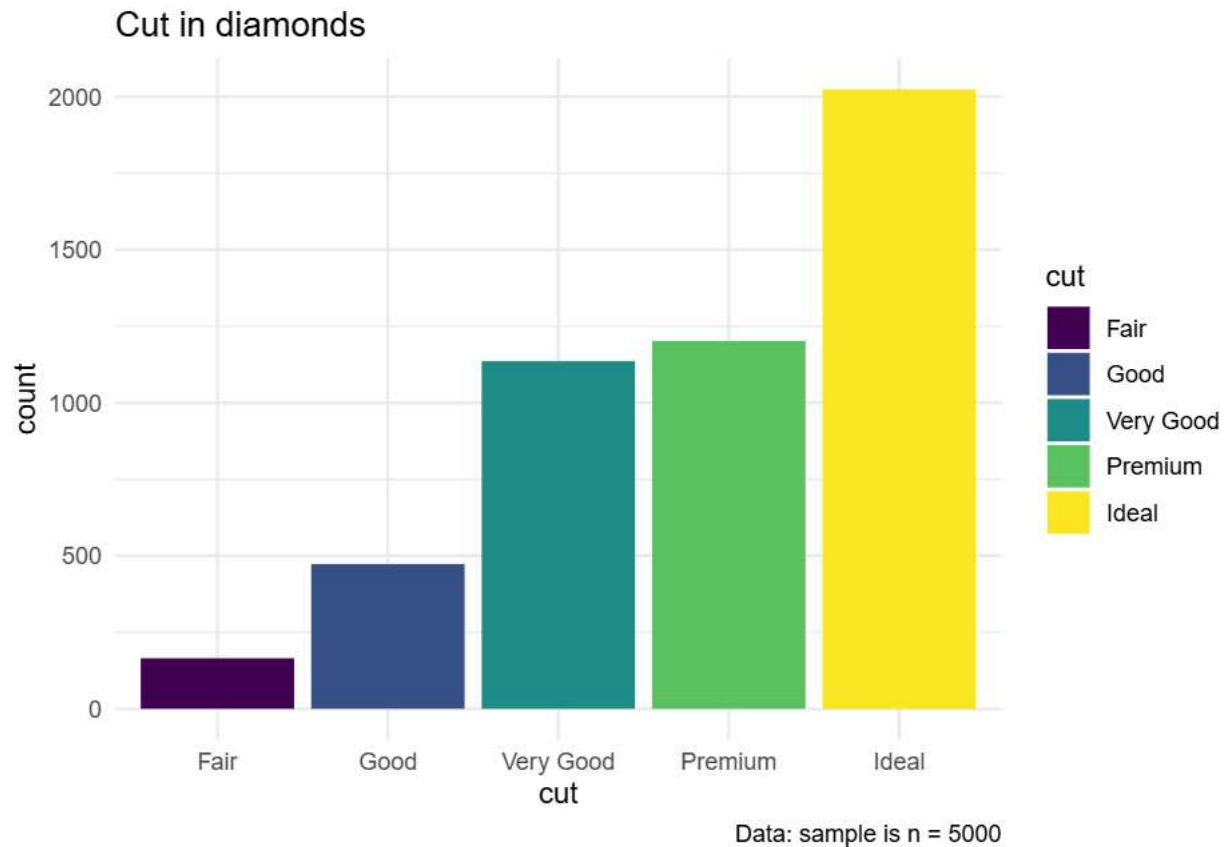
```
##      carat      cut      color      clarity      depth
## Min.   :0.2000  Fair      : 1610  D: 6775  SI1      :13065  Min.   :43.00
## 1st Qu.:0.4000  Good      : 4906  E: 9797  VS2      :12258  1st Qu.:61.00
## Median :0.7000  Very Good:12082  F: 9542  SI2      : 9194  Median :61.80
## Mean   :0.7979  Premium   :13791  G:11292  VS1      : 8171  Mean   :61.75
## 3rd Qu.:1.0400  Ideal     :21551  H: 8304  VVS2     : 5066  3rd Qu.:62.50
## Max.   :5.0100                I: 5422  VVS1     : 3655  Max.   :79.00
##                                J: 2808  (Other): 2531
##      table      price      x      y
## Min.   :43.00  Min.   : 326  Min.   : 0.000  Min.   : 0.000
## 1st Qu.:56.00  1st Qu.: 950  1st Qu.: 4.710  1st Qu.: 4.720
## Median :57.00  Median :2401  Median : 5.700  Median : 5.710
## Mean   :57.46  Mean   :3933  Mean   : 5.731  Mean   : 5.735
## 3rd Qu.:59.00  3rd Qu.:5324  3rd Qu.: 6.540  3rd Qu.: 6.540
## Max.   :95.00  Max.   :18823  Max.   :10.740  Max.   :58.900
##
##      z
## Min.   : 0.000
## 1st Qu.: 2.910
## Median : 3.530
## Mean   : 3.539
## 3rd Qu.: 4.040
## Max.   :31.800
##
```

Prepare data & Plot

Plot 1: diamonds sample(n=5000)

```
## sample data
set.seed(42)
diamonds_test1 <- diamonds %>%
  sample_n(5000)

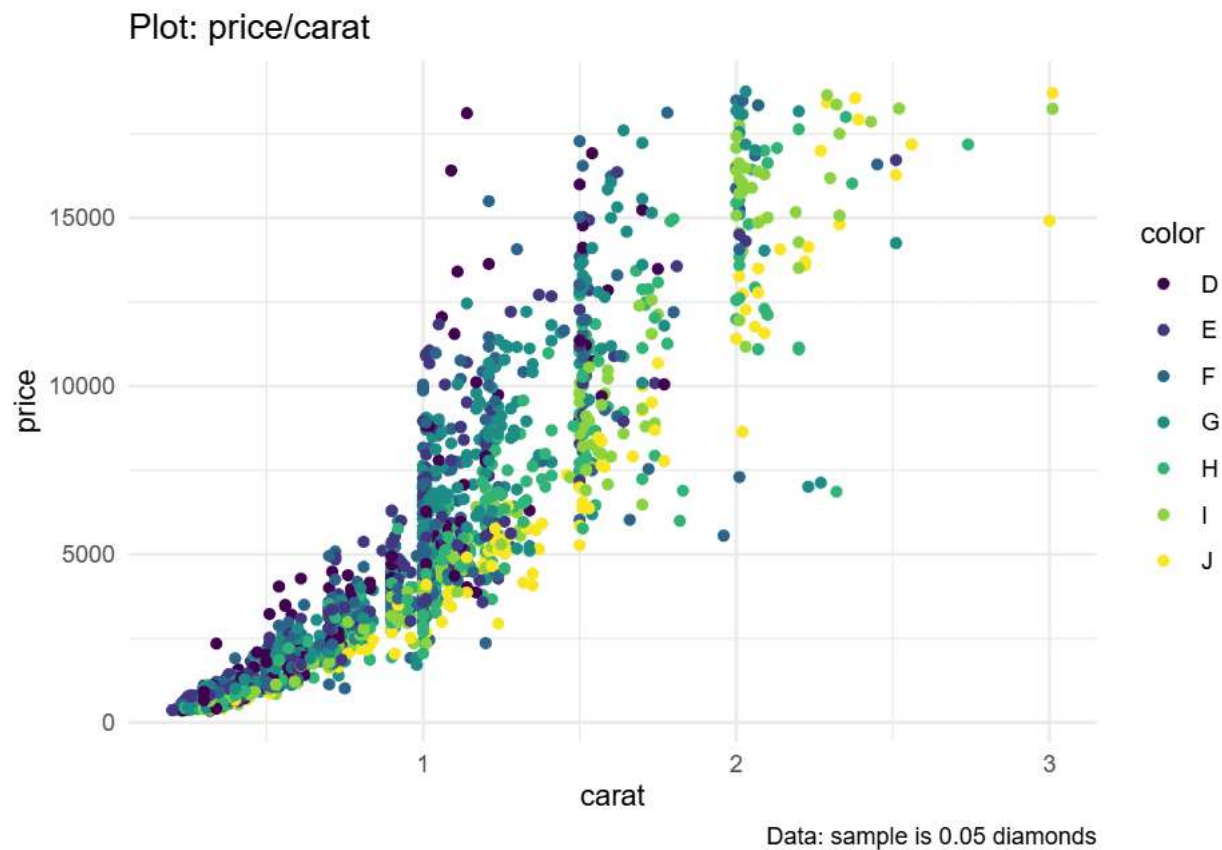
## plot
ggplot(diamonds_test1,
       aes(cut, fill = cut)) +
  geom_bar() +
  theme_minimal() +
  labs(
    title = "Cut in diamonds ",
    caption = "Data: sample is n = 5000",
    y = "count",
    x = "cut"
  )
```



Plot 2: diamonds ample_frac(0.05)

```
## sample data
set.seed(42)
diamonds_test2 <- diamonds %>%
  sample_frac(0.05)

## plot
ggplot(diamonds_test2,
  aes(carat, price, col=color)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Plot: price/carat ",
    caption = "Data: sample is 0.05 diamonds",
    y = "price",
    x = "carat"
  )
)
```



```
ggplot(data = mtcars, aes(x = mpg,y = wt)) +
  geom_point() +
  geom_smooth(method="lm") +
  theme_minimal() +
  labs(
    title = "Plot: mpg/wt ",
    caption = "Data: mtcars",
    y = "wt: Weight (1000 lbs)",
  )
```

```
x = "mpg: Miles/(US) gallon"  
)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Plot: mpg/wt

